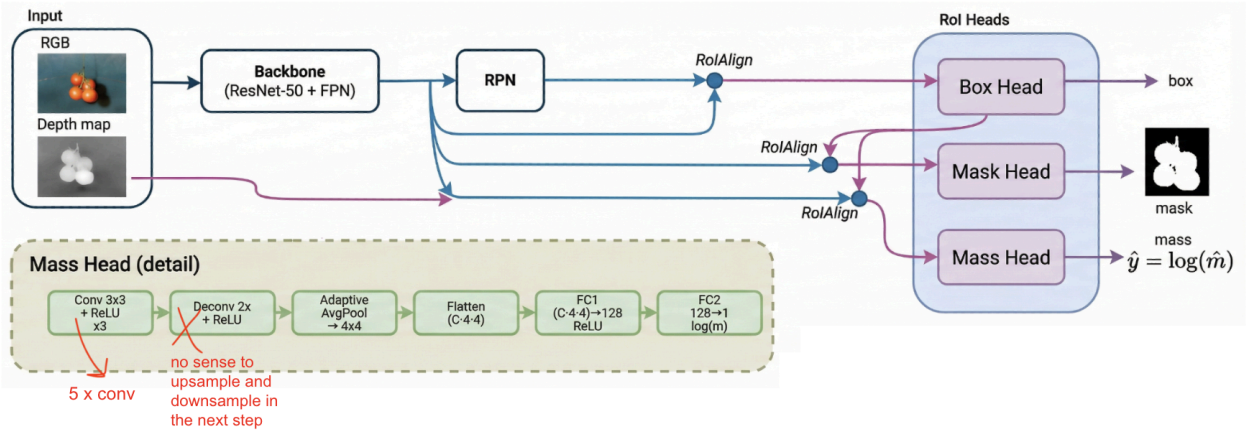
25.5 update

Model/(mass head) architecture



Experiments:

- Replace deconv in mass head with conv -> 5 x conv layers
- Experiments with 4x4 AvgPool vs 7x7 AvgPool (to sustain more spatial info)
- With vs without late fusion (no explicit depth map input to mass head)

Conclusion: all these variants have very small effects on final results < 2% MAPE

- Late fusion contributes almost nothing, difference ~ 0.3% MAPE
- Tried RGB dropout on the input -> depth channel gradients increase during training, but predictions don't change correspondingly. The model emits near-mean predictions, rather than actually using the depth.

Results from final RGB-D model

1.During training on last eval:

- MAPE = 14.3%
- R^2 = 0.86
- MAE (kg) = 0.113

2.Inference on train/val/test

Split	MAE (g)	MAPE	bias (g)	Worst
Train	31	**5.5 %**	-22	233
Val	119	**14.4 %**	-101	359
Test	114	**19.0 %**	-64	431

3.Depth effect tests

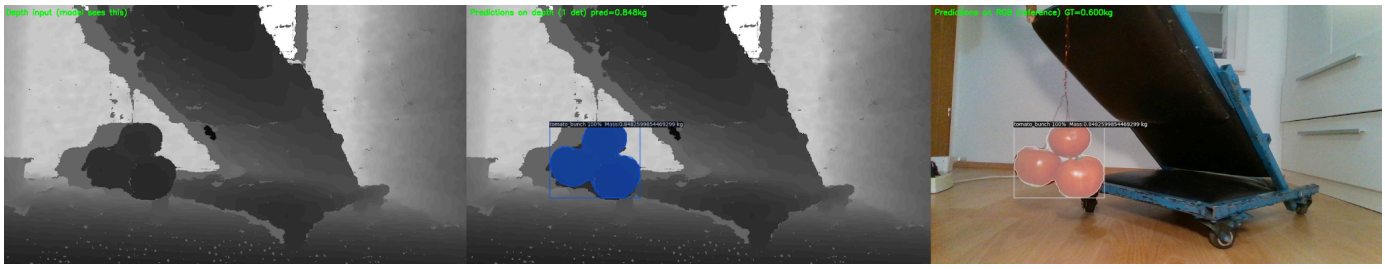
Variant	Train MAPE	Val MAPE	Test MAPE	Test MAE
1. real depth	5.5 %	14.4 %	19.0 %	114 g
2. depth/1000	5.4 %	14.3 %	19.0 %	114 g
3. zeros	5.4 %	14.3 %	19.0 %	114 g
4. 5000 mm constant	**5.0 %**	**13.0 %**	18.8 %	110 g

Conclusions:

- Model is depth-blind?

Depth-only model

- RGB channels: depth globally normalized to 0-255 - grayscale images
- Depth channel: raw depth in mm



Results compared to RGB-D model:

Metric	RGB-D best	Depth-only
Train `loss_mass` (final)	0.0029	**0.0017**
Val `loss_mass` (final, mean)	0.0379	0.0379
Val MAPE (mean over evals)	15.5 %	15.6 %
Val MAPE (last eval)	14.3 %	14.8 %
Val R^2 (last eval)	0.86	0.84
`bbox/AP`	92	88 (−4)
`segm/AP`	98	87 (−11)

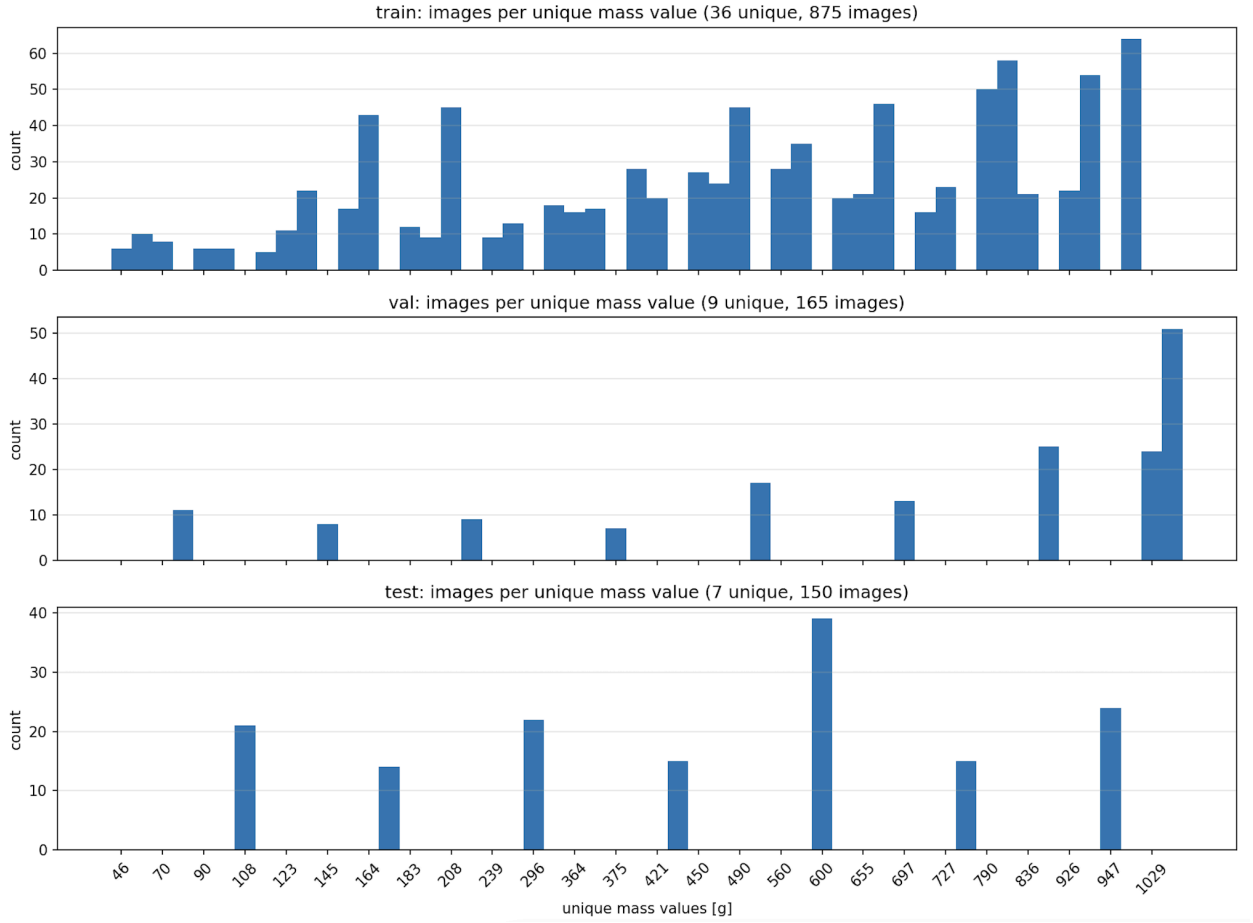
Split	RGB-D MAPE	Depth-only MAPE	RGB-D MAE	Depth-only MAE
Train	5.5 %	**3.4 %**	31 g	**18 g**
Val	14.4 %	**13.6 %**	119 g	**111 g**
Test	19.0 %	**14.1 %**	114 g	**73 g**

Depth test - same results

Conclusion/questions/thoughts:

- RGB pathway contributes nothing for mass estimation? (in means of color and lightning). Geometry signal is present in both depth and rgb. Mass is estimated from the geometry signal in the first 3 channels?
- The current RGB-D architecture has a structural limitation: depth as a single channel alongside many RGB / FPN channels is too small a signal-fraction to develop

Data analysis

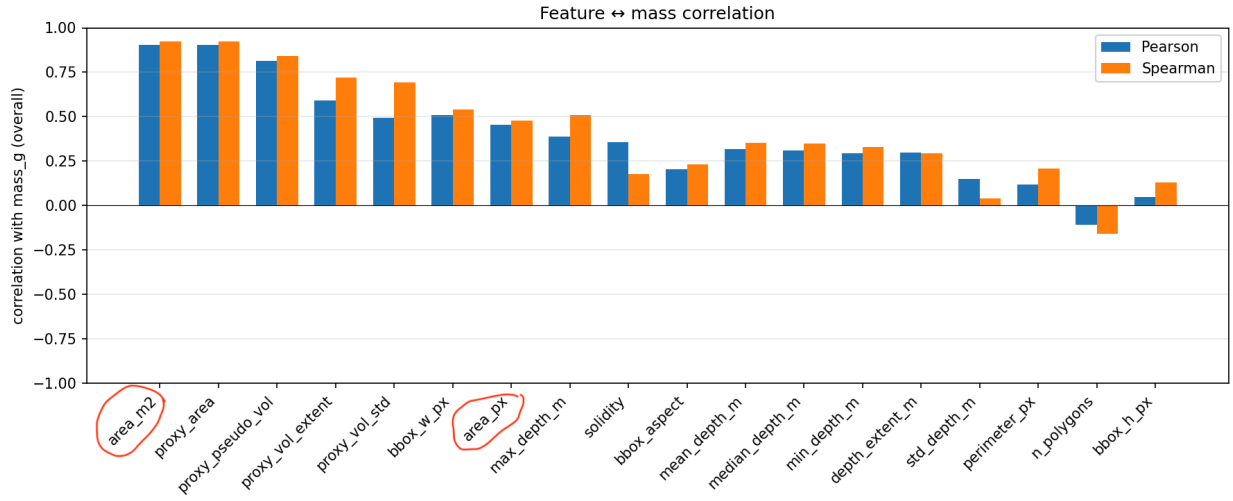


Dataset split by unique clusters:

- Train: 875 images, 36 unique mass values (mean 548.1 g, median 560.0 g, std 289.6 g)
- Val: 165 images, 9 unique mass values (mean 785.3 g, median 857.0 g, std 373.2 g)
- Test: 150 images, 7 unique mass values (mean 501.0 g, median 600.0 g, std 283.1 g)
- 1190 images, but 52 unique mass values
- Results sensitive to specific held-out mass groups

Idea: feature analysis to understand whether the dataset actually contains a learnable relationship between the RGB-D images and the GT mass values. To answer is there useful depth/geometry information in the dataset at all.

- Most important feature **area_m2** - visible metric area of the cluster using GT masks, depth map + cam intrinsics / area_m2 = $\Sigma Z^2 / (fx \cdot fy)$. Basically a depth-corrected version of the mask area
- Conclusion: Depth is not useful mainly as a standalone feature. Depth becomes useful when combined with the segmentation mask to recover metric object size?



1.6 update

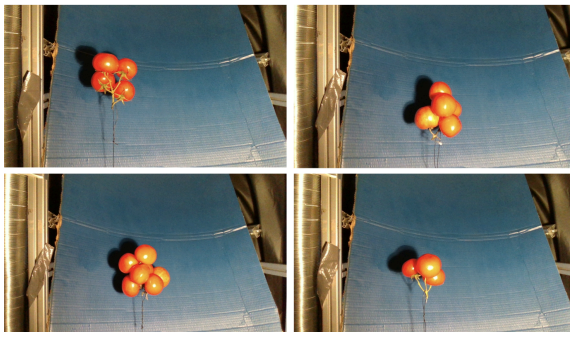
1. Input normalization scale

- Checked model input values after normalization:
  - rgb channels std ~ 57
  - depth channel std ~ 1
  - **why**: detectron2 convention - pretrained Mask R-CNN weights were trained that way (only mean-subtracted), the convolution filters expect inputs on that scale
  - **problem**: depth arrives 57x smaller, gradient following depth input weights is tiny, model barely learns to use depth
  - **try to fix**: set pixel_std for normalization of rgb channels to ~ 57 (standard ImageNet values), now all channels have std ~ 1
  - **result**: model adapts pre-trained filters during training, depth becomes readable

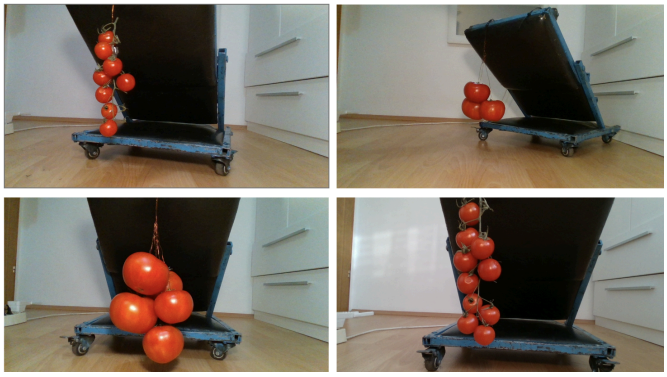
2. Dataset split

- 2 capture sessions: at home + at lab
- capture session at home:
  - includes most distance variations
  - distance range in captures for single tomato cluster: ~ 0.337 - 1.3 m
  - mass range 165 - 1160 g
- capture session in lab:
  - distance variation for single cluster: ~ 0.472 - 0.685 m
  - mass range 186 - 1029 g
  - total images: 416, unique clusters 21
- **idea**: Train model only on home dataset to force the distance variation to be learned + have a test dataset of completely unseen data (captures different rgb conditions than the train dataset)
- **new train dataset:** total images: 573, unique clusters 14
  - train: 383 images; 10 clusters
  - val: 190 images; 4 clusters (208, 490, 670, 940) g

Lab dataset

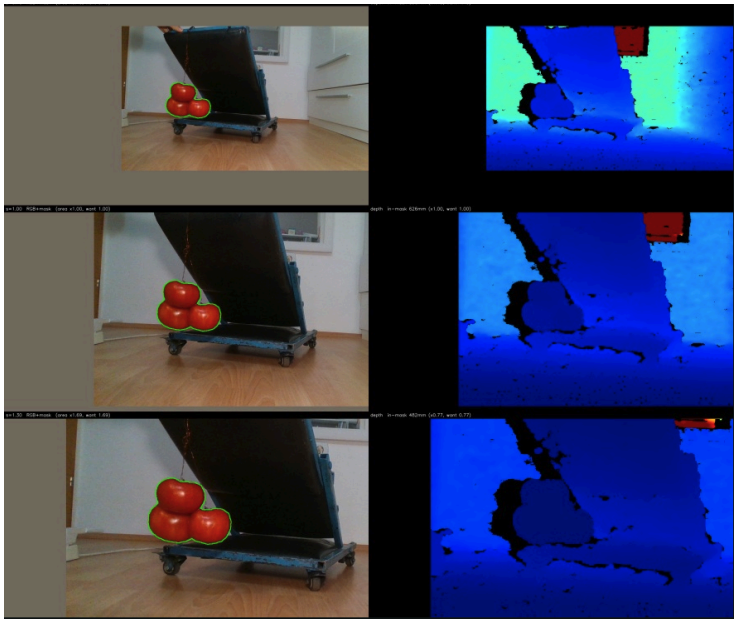


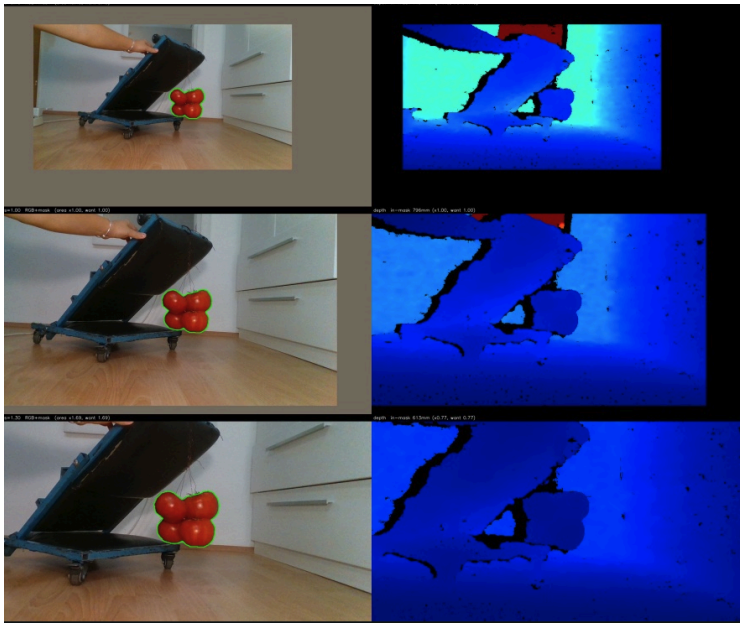
Home dataset

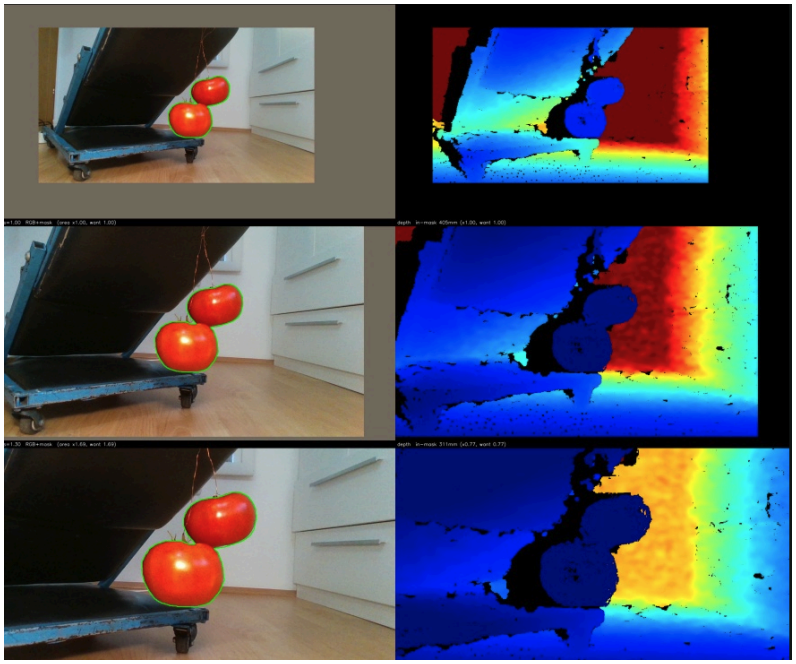


3. Distance/zoom augmentation

- Trying to add more distance variation into the dataset so model doesn't learn: bigger blob -> heavier mass
- **How it is done**: during training random zoom factor is picked (0.7 - 1.3) and simulates camera moving closer/further
    - resize pixels by s -> size becomes x s
    - scale depth values by 1/s
    - keep original size centered cluster - crop image if zoomed in; pad if zoomed out
- Example images:







4. Trained model variations

Main groups:

Model group	Train data	Input	Late Fusion	Norm	Augmentation	Purpose
1. RGB-D baseline	lab + home	RGBD	yes/no	detectron2 default	horizontal flip	baseline
2.Depth visual	lab + home	ch 0-2 depth grayscale; ch 3 depth in mm	yes/no	default	flip	test geometry without RGB
3.RGB-D	home	RGBD	no	RGB + D std (same scale)	flip	Test normalization
4.RGB-D	home	RGBD	no	default/RGB + D std (same scale)	RGB dropout 0.5	Reduce RGB shortcut
5.RGB-D	home	RGBD	no	RGB + D std (same scale)	Zoom + RGB dropout 0.5	Force more distance invariance

In-domain (on val dataset; same capture session as train, different clusters) **MAPE** results:

Model group	Real RGB-D input	RGB-only input	Depth-only input	Thoughts
4.1	8.5%	8.5%	53%	Mostly rgb driven for mass estimation
4.2	12%	11.9%	13.6%	Much better results with normalization fix
5.	10.3%	10.9%	11%	Most consistent results across the 3 input variations
5. + additional brightness/contrast color augmentation	14.6%	13.1%	14.5%	Lower results on in-domain images than 5.

Conclusions for in-domain results:

- Most models achieve good in-domain MAPE, however RGB-only vs RGB-D input gives the ~ same result -> when RGB is present, model decides based on it
- Models trained with no rgb dropout don't detect anything with depth-only input

Cross-session generalization (inference on lab test dataset/ completely different session than train). **MAPE** results:

Model	Real RGB-D input	RGB-only input	Depth-only input
4.1	50.4%	50.6%	43.9%
4.2	70.7%	70.1%	37.5%
5.	66.8%	65.5%	35.8%
5. + additional brightness/contrast color augmentation	58.7%	55.5%	33.9%

- Models trained on home dataset perform well in-domain but generalize poorly to the lab dataset.

- RGB-D and RGB-only tests show poorer results for mass estimation than depth-only input tests.
- RGB appearance/background/lightning cues are session-specific shortcuts
- Depth information transfers better, but still not good enough for robust generalization